



La Grande Ecole de l'IA & de la Data

Paris Île-de-France · Nice Côte d'Azur

PGE Final Project (PFE)

Research Methods

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Attention !

You are not asked to provide a summary of a subject such as:

- **Example 1: Methodologies for handling missing data in statistical analyses**
- **Example 2: A comparative study of data visualization tools: strengths, limitations and applications**

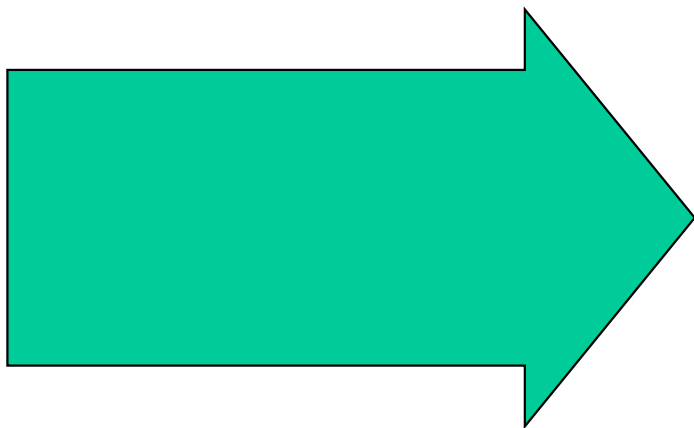
Attention !
**we are not asking you for an
internship report**

- **An internship report is a description focused on the mission(s) carried out during the internship**
- **This is another approach...**

Objective of PFE

The thesis = A study and research work relating to a “problem” on AI and Data science

- Demonstrate mastery of the skills acquired during the PGE course
- Solve a concrete problem related to AI and data science
- Integrate theoretical and practical approaches on AI and data science



- **This involves answering one or more questions**
- **Questions related to a specific topic/problem**
- **More general questions**

The Applied Research Report / PFE

- A real action project that addresses a advanced technological problem in AI and data-science and leads to the development of concrete proposals allowing you to apply the concepts, methods and tools you have learned during your PGE studies.

Possible Pathways

- **1. You choose a project that is aligned with the company project you are doing**
- **2. You choose a project relevant to a topic that you want to pursue in future**

Global approach

Overall progress

- **Delve deeper into the topic that interests you and check its relevance**
- **Reading significant professional or more academic publications on the topic you have chosen + possibly interviews with professionals**
- **Pose the problem**
- **Choose one or more operational (or more general) questions**
- **Choose an analysis angle**
- **Choose the method(s) for collecting information that will allow you to answer the question asked.**
- **Interact with senior researchers in the chosen field for guidance.**
- **Collect information or design an AI/Data solution**
- **Analyze the information collected to answer the question(s) asked or analyze the results of your solution**
- **Take a step back and do a critical analysis**
- **Put your results into perspective with the existing literature on the subject**

The general approach

The PFE thesis must:

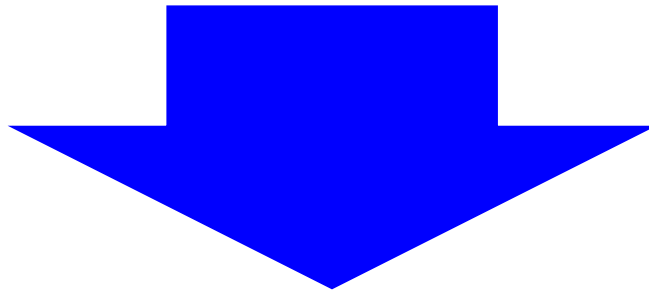
- **Respect a rigorous scientific approach (methodology)**
- **Be informed (by in-depth literature readings)**
- **Be technically strong**
- **Theoretical reflection must be supported with data from the field.**
- **The construction of the research object will require you to go back and forth between literature and your work to refine the ideas.**

Choice of subject

- **The subject of the PFE must result above all from the student's readings, meetings, practices, tastes and reflections.**
- **The process of choosing the subject can also result from the work to be done in the company, the opinion of the apprenticeship tutor and the student's career aspirations.**
- **It would be interesting to have a good harmony between the work done in the company and the subject of the dissertation. The greater the synergies, the easier the work, the better supported by the company because more expected and therefore helped.**

What is a project problem ?

- **The problem is not limited to a general theme or subject.**



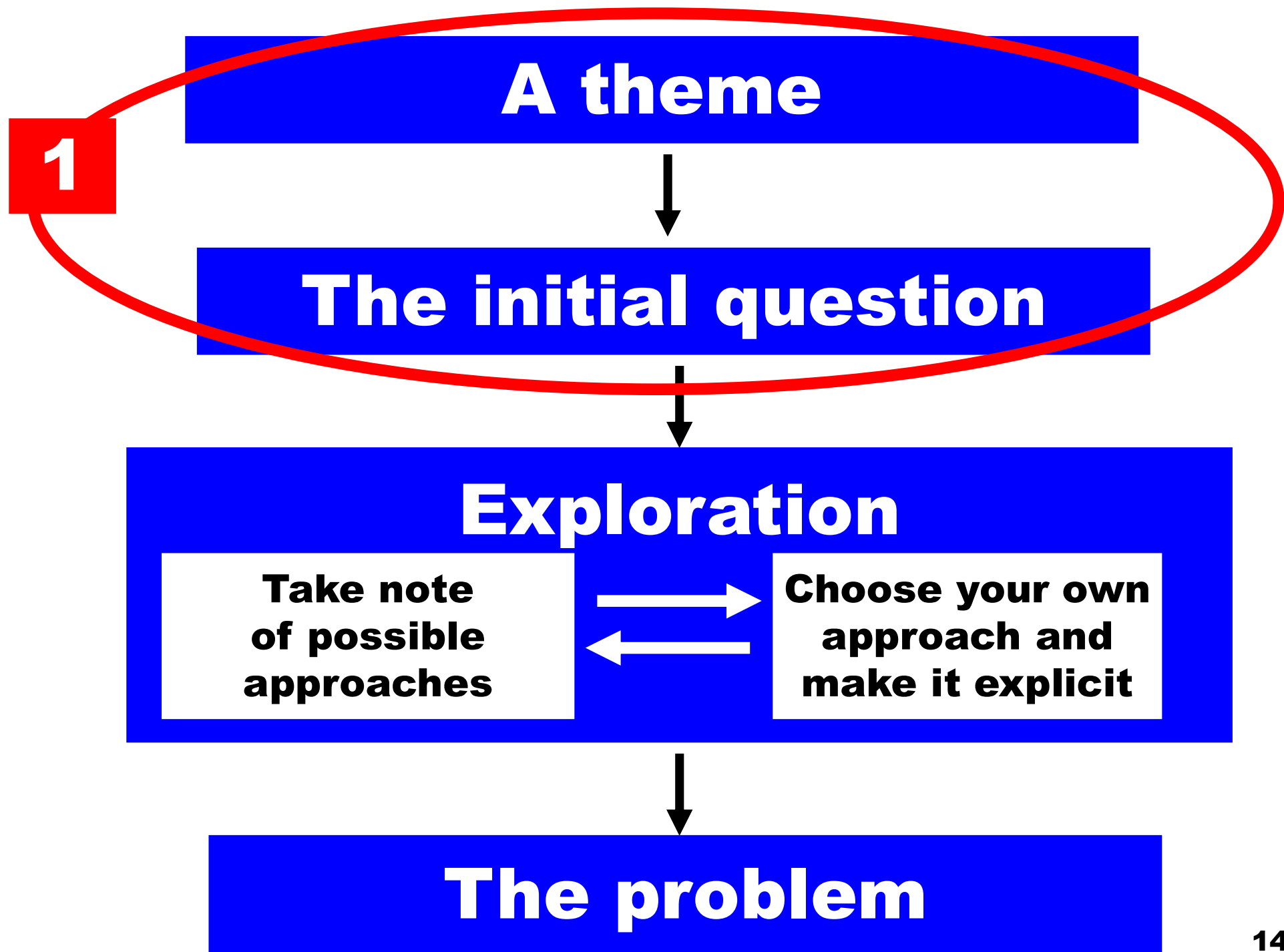
- **This is a specific question in a specific context (it is not the research question)**

What is a project problem?

- **The problem is the outcome of the choice of the subject: it allows both to define and justify the way in which the subject will be treated.**
- **The problem explains the chosen project and its workflow. In this context, it is necessary to ask a main question which must be central and essential in relation to the subject, that you show why it is interesting and on what it is based.**

How to start the research?

- **1- Start with a subject: generally related to your mission in the company OR your preferred field for future work**
- **2- You do an literature/documentary research on the subject**
- **3- From all the information you have observed/read, you choose a question**
- **Try to specify your question(s) as best as possible**
- **After defining the research question, you will be able to develop a timeline for the project activities (e.g. for completing literature survey, formulating the problem, getting the data, developing the methodology, getting the results, creating project report and presentation).**



A theme



The initial question



Exploration

2

**Take note
of possible
approaches**



**Choose your own
approach and
make it explicit**



The problem

The exploration

**Take note
of existing
approaches**



**Choose your own
approach and
explain why
it is interesting**

■ 1 : Identify possible approaches

- Take note of the various possible approaches to the problem and understand their specificities
- Select the approach you want to develop
- Making choices means giving up certain approaches
- It is about answering the question: how am I going to approach this problem?
- What is the object of my study?
- What type of perspective do I have on this object?
- Are there available datasets ?

Strategies

Where to find information about your project topic at this stage?

- **Academic literature**
 - Articles in the specialized professional press
 - General databases (google scholar, scopus)
 - "Scientific" works
 - Scientific articles / academic journals
 - Academic databases (EBSCO - Science Direct)...
- **Exploratory interviews: Search for interviews by famous researchers/academicians, industry founders in your field who talk about latest developments.**
- **Go through company pages who are working in your field of choice to study the domains of practical interest**



Date indifférente

Depuis 2024

Depuis 2023

Depuis 2020

Période spécifique...

Trier par pertinence

Trier par date

Toutes les langues

Recherche dans les
pages en Français

Tous les types

Articles de revue

☐ inclure les brevets

☒ inclure les citations

☒ Créer l'alerte

A survey on model-based reinforcement learning

[FM Luo](#), [T Xu](#), [H Lai](#), [XH Chen](#), [W Zhang](#)... - Science China Information ..., 2024 - Springer

... Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor. In: Proceedings of the 35th International Conference on Machine Learning, 2018. 1856–...

☆ Enregistrer Citer Cité 39 fois Autres articles Les 2 versions

A definition of continual reinforcement learning

[D Abel](#), [A Barreto](#), [B Van Roy](#)... - Advances in ..., 2024 - proceedings.neurips.cc

... reinforcement learning as the setting in which the best agents are all continual learning ... of multi-task reinforcement learning and continual supervised learning are special cases of ...

☆ Enregistrer Citer Cité 7 fois Autres articles Les 7 versions

Reflexion: Language agents with verbal reinforcement learning

[N Shinn](#), [F Cassano](#), [A Gopinath](#)... - Advances in ..., 2024 - proceedings.neurips.cc

... On the other hand, reinforcement learning has suffered from its black-box policy and ... Our proposed “verbal” reinforcement learning might address some of the issues and turn ...

☆ Enregistrer Citer Cité 216 fois Autres articles Les 2 versions

Supervised pretraining can learn in-context reinforcement learning

[J Lee](#), [A Xie](#), [A Pacchiano](#), [Y Chandak](#)... - Advances in ..., 2024 - proceedings.neurips.cc

... and reinforcement learning, our approach takes inspiration from general in-context learning, ... surprisingly lead to a model that exhibits meta-learning capabilities [1]. Recently, there has ...

☆ Enregistrer Citer Cité 13 fois Autres articles

3 Main Types of Scientific Articles (Academic Journals)

**Review of
the literature**

**Takes stock of
research work
on a theme**

**Research
without
empirical
data**

**Proposes
research
hypotheses
based
on
a
theoretical
reflection**

**Research
with
empirical
data**

**Technical
research
article
with data
and methods**

Material to prepare for the April session (Mémoire/Report)

- **Research question (3 formulation proposals)**
- **Research problem (2 to 3 pages) – Version 1**
- **10 bibliographic references including 5 scientific articles**
- **Summary presentation of the planned empirical part (1 to 2 pages):**
 - **The project (what data, what methods, what expected results, type(s) of deliverables)**
 - **OR: The empirical study: Why?, What empirical methods (type of data collected, analysis methods, Hypotheses)**
- **You can compile this material in either French or English**

Important dates

- **February session: Finish identifying a thematic area, a potential topic, search for relevant data sources, possible technical methodologies**
- **March session: Complete Initial literature review, collection of dataset, identification of methodologies**
- **April: Present the Memoire/Report- Phase 1 (see last slide), discuss progress, identify problems, assigning of specific professors for projects**
- **July: Results and analysis work to be finalized**
- **September: Final PFE presentation**